

FagusED · Physics
01 · Motion, forces and energy

How is average speed calculated?

PHY-0001

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

How is acceleration calculated?

PHY-0002

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

What does the gradient of a distance-time graph represent?

PHY-0003

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

What does the gradient of a velocity-time graph represent?

PHY-0004

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

What happens when the resultant force on an object is zero?

PHY-0005

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

State the relationship between force, mass and acceleration.

PHY-0006

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

How is weight different from mass?

PHY-0007

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

How is momentum calculated?

PHY-0008

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

How is work done calculated for a force along the motion?

PHY-0009

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

What is the energy of motion called and how is it calculated?

PHY-0010

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

What is gravitational potential energy gained near Earth?

PHY-0011

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

How do you calculate efficiency?

PHY-0012

PREGUNTA

FagusED · Physics
02 · Thermal physics

How does a gas create pressure on a container wall?

PHY-0013

PREGUNTA

FagusED · Physics
02 · Thermal physics

Why does pressure increase when a sealed gas is heated at constant volume?

PHY-0014

PREGUNTA

FagusED · Physics
02 · Thermal physics

What is internal energy?

PHY-0015

PREGUNTA

FagusED · Physics
02 · Thermal physics

What is specific heat capacity?

PHY-0016

PREGUNTA

FagusED · Physics
02 · Thermal physics

What is specific latent heat?

PHY-0017

PREGUNTA

FagusED · Physics
02 · Thermal physics

Why is temperature constant during a pure substance's melting?

PHY-0018

PREGUNTA

FagusED · Physics
02 · Thermal physics

How is heat transferred by conduction?

PHY-0019

PREGUNTA

FagusED · Physics
02 · Thermal physics

How is heat transferred by convection?

PHY-0020

PREGUNTA

FagusED · Physics
02 · Thermal physics

Which surfaces absorb and emit infrared radiation best?

PHY-0021

PREGUNTA

FagusED · Physics
01 · Motion, forces and energy

Speed. A steeper line means a greater speed.

PHY-0003

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Acceleration = change in velocity ÷ time taken; it has units m/s².

PHY-0002

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Average speed = total distance ÷ total time; include a suitable speed unit.

PHY-0001

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Resultant force = mass × acceleration ($F = ma$).

PHY-0006

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Its velocity stays constant: it may remain at rest or continue in a straight line at constant speed.

PHY-0005

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Acceleration; the signed area between the graph and time axis gives displacement.

PHY-0004

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Work done = force × distance moved in the direction of that force.

PHY-0009

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Momentum $p = mv$; in a closed system total momentum is conserved.

PHY-0008

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Mass measures inertia in kilograms; weight is gravitational force, $W = mg$, in newtons.

PHY-0007

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Efficiency = useful energy output ÷ total energy input × 100%, or useful power output ÷ total power input × 100%.

PHY-0012

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Change in GPE = $mg\Delta h$.

PHY-0011

RESPUESTA

FagusED · Physics
01 · Motion, forces and energy

Kinetic energy = $\frac{1}{2}mv^2$.

PHY-0010

RESPUESTA

FagusED · Physics
02 · Thermal physics

The sum of particles' random kinetic energies and their potential energies from particle interactions.

PHY-0015

RESPUESTA

FagusED · Physics
02 · Thermal physics

Particles gain kinetic energy; collisions with the walls become more frequent and cause larger momentum changes.

PHY-0014

RESPUESTA

FagusED · Physics
02 · Thermal physics

Gas particles collide with the wall, changing momentum and exerting force over its area.

PHY-0013

RESPUESTA

FagusED · Physics
02 · Thermal physics

Added energy weakens attractions and increases potential energy, rather than average kinetic energy.

PHY-0018

RESPUESTA

FagusED · Physics
02 · Thermal physics

Energy per kilogram needed for a change of state at constant temperature; $E = mL$.

PHY-0017

RESPUESTA

FagusED · Physics
02 · Thermal physics

Energy required to raise the temperature of 1 kg of a substance by 1 °C; $E = mc\Delta T$.

PHY-0016

RESPUESTA

FagusED · Physics
02 · Thermal physics

Dark, matt surfaces; shiny, light surfaces are comparatively poor emitters and absorbers.

PHY-0021

RESPUESTA

FagusED · Physics
02 · Thermal physics

Warm fluid expands, becomes less dense and rises while cooler fluid sinks, creating a circulation.

PHY-0020

RESPUESTA

FagusED · Physics
02 · Thermal physics

Energetic particles transfer energy by collisions; free electrons also contribute in metals.

PHY-0019

RESPUESTA

FagusED · Physics
02 · Thermal physics

How does evaporation cool a liquid?

PHY-0022

PREGUNTA

FagusED · Physics
03 · Waves

Explain the main idea of wave quantities.

PHY-0023

PREGUNTA

FagusED · Physics
03 · Waves

Explain the main idea of transverse and longitudinal waves.

PHY-0024

PREGUNTA

FagusED · Physics
03 · Waves

Which quantity is measured in hertz?

- A) frequency
- B) wavelength
- C) amplitude
- D) wave speed

PHY-0025

PREGUNTA

FagusED · Physics
03 · Waves

A wave has frequency 4.0 Hz and wavelength 0.50 m. What is its speed?

- A) 0.125 m/s
- B) 2.0 m/s
- C) 4.5 m/s
- D) 8.0 m/s

PHY-0026

PREGUNTA

FagusED · Physics
03 · Waves

Which wave is longitudinal?

- A) light in vacuum
- B) sound in air
- C) radio wave
- D) wave on a string

PHY-0027

PREGUNTA

FagusED · Physics
03 · Waves

When light reflects from a mirror, the angle of reflection is measured from the

- A) mirror surface
- B) incident ray
- C) normal
- D) reflected ray

PHY-0028

PREGUNTA

FagusED · Physics
03 · Waves

Light enters glass from air and slows down. It bends

- A) away from the normal
- B) towards the normal
- C) along the boundary
- D) without changing direction

PHY-0029

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

Explain the main idea of charge, current and potential difference.

PHY-0030

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

Explain the main idea of resistance and current-voltage behaviour.

PHY-0031

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

Which equation defines current?

- A) $I = Q/t$
- B) $I = Qt$
- C) $I = VQ$
- D) $I = E/t$

PHY-0032

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

A 6.0 V resistor carries 0.50 A. Its resistance is

- A) 3.0 Ω
- B) 12 Ω
- C) 30 Ω
- D) 0.083 Ω

PHY-0033

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

What is the current at every point in a series circuit?

- A) always zero
- B) the same
- C) different at each resistor
- D) equal to voltage

PHY-0034

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

Which quantity is the same across parallel branches?

- A) current
- B) charge
- C) potential difference
- D) resistance

PHY-0035

PREGUNTA

FagusED · Physics
04 · Electricity and magnetism

A 230 V appliance draws 4.0 A. Its power is

- A) 57.5 W
- B) 226 W
- C) 234 W
- D) 920 W

PHY-0036

PREGUNTA

FagusED · Physics
03 · Waves

In a transverse wave, oscillations are perpendicular to energy transfer; examples include electromagnetic waves and waves on a stretched string. In a longitudinal wave, oscillations are parallel to energy transfer and produce compressions and rarefactions; sound in air is longitudinal.

PHY-0024

RESPUESTA

FagusED · Physics
03 · Waves

A wave transfers energy without a net transfer of matter. Amplitude is maximum displacement from equilibrium; wavelength is distance between corresponding points; frequency is oscillations per second; period is time for one oscillation.

PHY-0023

RESPUESTA

FagusED · Physics
02 · Thermal physics

Higher-energy particles escape at the surface, lowering the average kinetic energy of those remaining.

PHY-0022

RESPUESTA

FagusED · Physics
03 · Waves

B) sound in air

PHY-0027

RESPUESTA

FagusED · Physics
03 · Waves

B) 2.0 m/s

PHY-0026

RESPUESTA

FagusED · Physics
03 · Waves

A) frequency

PHY-0025

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

Electric charge is positive or negative. Current is the rate of flow of charge; conventional current is from positive to negative outside a source. Potential difference is energy transferred per unit charge, while e.m.f. is energy supplied per unit charge by a source.

PHY-0030

RESPUESTA

FagusED · Physics
03 · Waves

B) towards the normal

PHY-0029

RESPUESTA

FagusED · Physics
03 · Waves

C) normal

PHY-0028

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

B) 12 Ω

PHY-0033

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

A) $I=Q/t$

PHY-0032

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

Resistance opposes current. For an ohmic conductor at constant temperature, current is proportional to voltage. A filament lamp heats as current rises, increasing resistance; a diode conducts mainly in one direction.

PHY-0031

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

D) 920 W

PHY-0036

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

C) potential difference

PHY-0035

RESPUESTA

FagusED · Physics
04 · Electricity and magnetism

B) the same

PHY-0034

RESPUESTA

FagusED · Physics
05 · Nuclear physics

Explain the main idea of atomic structure and nuclide notation.

PHY-0037

PREGUNTA

FagusED · Physics
05 · Nuclear physics

Explain the main idea of radioactive decay and detection.

PHY-0038

PREGUNTA

FagusED · Physics
05 · Nuclear physics

Which particles are in the nucleus?
A) protons and neutrons
B) protons and electrons
C) neutrons and electrons
D) electrons only

PHY-0039

PREGUNTA

FagusED · Physics
05 · Nuclear physics

Two isotopes of the same element have different numbers of
A) protons
B) electrons in neutral atoms
C) neutrons
D) nuclear charge

PHY-0040

PREGUNTA

FagusED · Physics
05 · Nuclear physics

What is random about radioactive decay?
A) the type of element
B) which nucleus decays and when
C) the existence of a nucleus
D) the charge of a proton

PHY-0041

PREGUNTA

FagusED · Physics
05 · Nuclear physics

Which radiation is most ionising?
A) alpha
B) beta
C) gamma
D) infrared

PHY-0042

PREGUNTA

FagusED · Physics
05 · Nuclear physics

Which material stops beta radiation effectively?
A) paper only
B) thin aluminium
C) air only
D) a magnetic field

PHY-0043

PREGUNTA

FagusED · Physics
06 · Space physics

Explain the main idea of the solar system and gravity.

PHY-0044

PREGUNTA

FagusED · Physics
06 · Space physics

Explain the main idea of the sun and energy.

PHY-0045

PREGUNTA

FagusED · Physics
06 · Space physics

What provides the centripetal force for a planet orbiting the Sun?
A) gravity
B) magnetism
C) air resistance
D) nuclear force

PHY-0046

PREGUNTA

FagusED · Physics
06 · Space physics

A planet completes an orbit of radius r in period T . Its orbital speed is
A) r/T
B) $2\pi r/T$
C) $T/2\pi r$
D) $2\pi T/r$

PHY-0047

PREGUNTA

FagusED · Physics
06 · Space physics

Why is an orbiting satellite accelerating at constant speed?
A) its mass changes
B) its direction changes
C) gravity is zero
D) its period is infinite

PHY-0048

PREGUNTA

FagusED · Physics
06 · Space physics

What process releases energy in the Sun?
A) fission of uranium
B) fusion of hydrogen
C) chemical combustion
D) radioactive beta decay only

PHY-0049

PREGUNTA

FagusED · Physics
06 · Space physics

Where do stars begin to form?
A) nebulae
B) planetary cores
C) empty vacuum
D) black-hole event horizons

PHY-0050

PREGUNTA

FagusED · Physics
05 · Nuclear physics

A) protons and neutrons

PHY-0039

RESPUESTA

FagusED · Physics
05 · Nuclear physics

Unstable nuclei decay randomly and spontaneously. Activity is decays per second in becquerels. Count rate includes background, which should be measured and subtracted. A Geiger-Müller tube detects ionising radiation.

PHY-0038

RESPUESTA

FagusED · Physics
05 · Nuclear physics

An atom has a tiny positive nucleus containing protons and neutrons, surrounded by electrons. Proton number Z identifies the element; nucleon number A is protons plus neutrons. Isotopes have the same Z but different neutron numbers.

PHY-0037

RESPUESTA

FagusED · Physics
05 · Nuclear physics

A) alpha

PHY-0042

RESPUESTA

FagusED · Physics
05 · Nuclear physics

B) which nucleus decays and when

PHY-0041

RESPUESTA

FagusED · Physics
05 · Nuclear physics

C) neutrons

PHY-0040

RESPUESTA

FagusED · Physics
06 · Space physics

The Sun is a medium-sized star containing most of the Solar System mass. Fusion of hydrogen nuclei releases energy. Electromagnetic radiation travels from the Sun through the vacuum; light takes about 8 minutes to reach Earth.

PHY-0045

RESPUESTA

FagusED · Physics
06 · Space physics

The Solar System contains the Sun, eight planets, dwarf planets, moons, asteroids and comets. Gravity provides the centripetal force for orbits. Orbital speed depends on orbital radius and period; closer planets generally move faster.

PHY-0044

RESPUESTA

FagusED · Physics
05 · Nuclear physics

B) thin aluminium

PHY-0043

RESPUESTA

FagusED · Physics
06 · Space physics

B) its direction changes

PHY-0048

RESPUESTA

FagusED · Physics
06 · Space physics

B) $2\pi r/T$

PHY-0047

RESPUESTA

FagusED · Physics
06 · Space physics

A) gravity

PHY-0046

RESPUESTA

FagusED · Physics
06 · Space physics

A) nebulae

PHY-0050

RESPUESTA

FagusED · Physics
06 · Space physics

B) fusion of hydrogen

PHY-0049

RESPUESTA